

Why Learned Feasibility Classification Fails Here

v11 — Ten diagnosed negative results, a baseline reality check, and a first win from augmenting CP-SAT

Motivation: Feasibility Classification, Directly

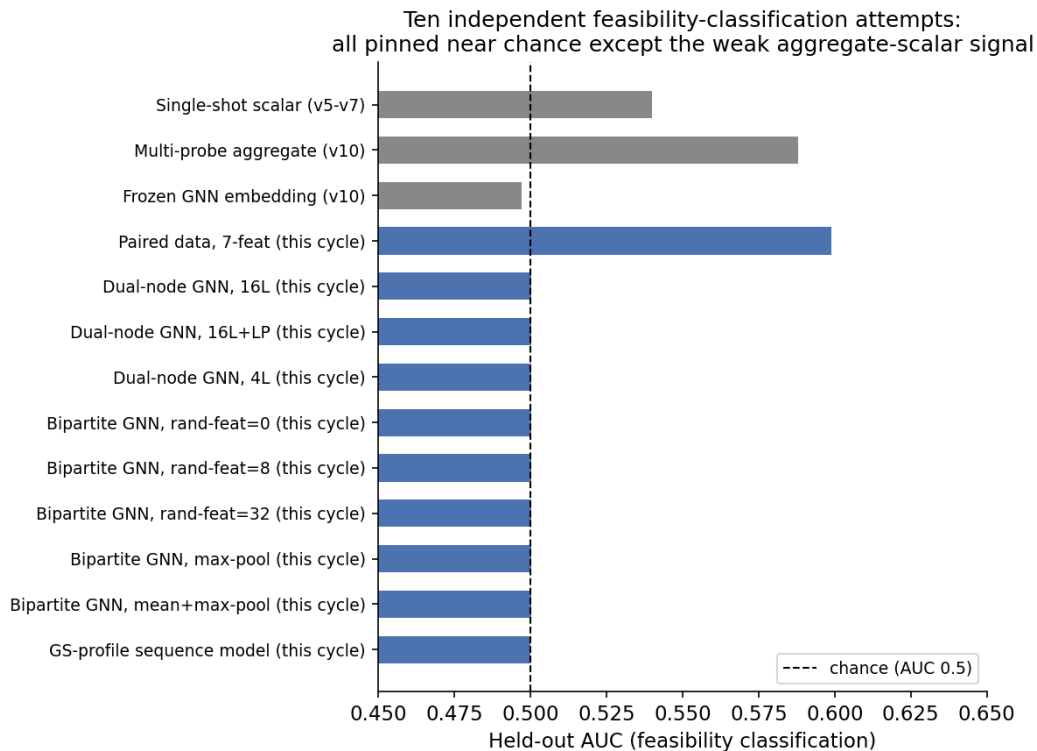
Solution construction set aside — classify feasibility (does a 0/1 solution exist) directly

Requirement A: inference faster than classical MILP/CP-SAT solvers

Requirement B: accuracy that survives zero-shot transfer to untrained sizes

This cycle tests both requirements head-on, against our own experiments and the literature

Ten Independent Negative Results



Ruling Out Optimization Bugs

Every negative result checked against a sanity task before being trusted

Dual-node GNN: loss 0.693→0.607 in 25 epochs on an easy continuous proxy

Bipartite GNN: loss 0.614→0.04 in 6 epochs on a pooling-preserved signal

→ Gradients and optimizer work fine. The failure is not a training bug.

Root Cause: Near-Miss Pairs Are Representationally Identical

A sequence model on the raw BKZ Gram-Schmidt profile also lands at AUC 0.5000

Diagnosis: 96.8% of 2,529 near-miss pairs are bit-identical across all 8 BKZ tries

b affects only 1 of $n+1$ basis rows — the rest are entirely determined by A (shared)

→ Not a training/capacity issue: the representation itself carries no information

Literature Grounding

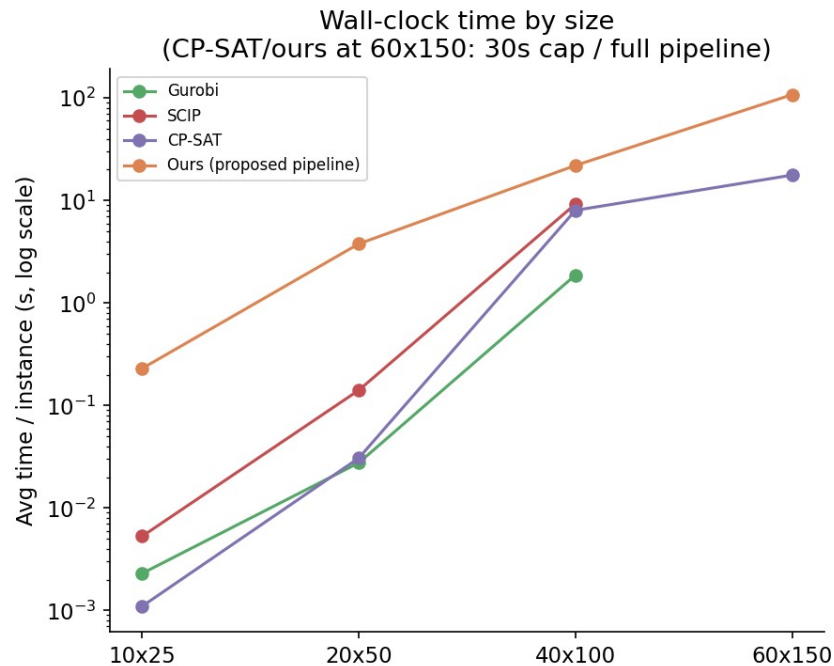
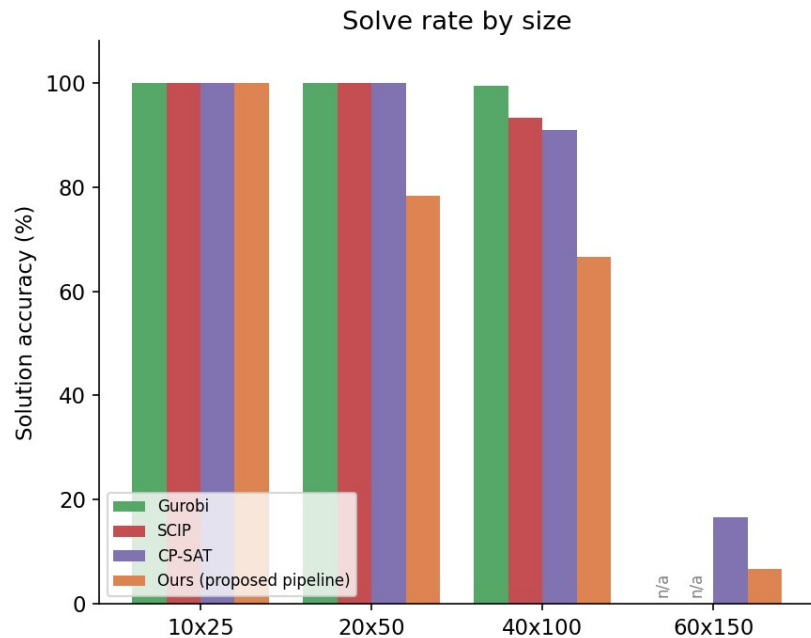
Chen et al. 2023 (ICLR): some feasible/infeasible MILP pairs are provably GNN-indistinguishable

→ Their fix (random node features) tested this cycle, no effect here (d=0/8/32 all 0.5001)

G4SATBench (TMLR 2024): SAT classifiers lose 15-25 accuracy points under 4-5x size transfer

RL whose output is a feasibility verdict: essentially absent from the literature surveyed

Baseline Reality Check



Accuracy & Time by Size

Size	Gurobi	SCIP	CP-SAT	Ours
10x25	100.0% / 0.0023s	100.0% / 0.0053s	100.0% / 0.0011s	100.0% / 0.23s
20x50	100.0% / 0.028s	100.0% / 0.141s	100.0% / 0.031s	78.25% / 3.78s
40x100	99.4% / 1.875s	93.2% / 9.24s	90.9% / 8.03s	66.67% / 22.0s
60x150	—	—	16.67% / 17.72s	6.67% / 107.95s

Baselines up to 200x faster and more accurate through 40x100. 60x150 tested for the first time this cycle -- hard for CP-SAT too (2/12), but still ahead of ours (2/30).

Augmenting, Not Replacing, CP-SAT

Variant	Sol. Acc.	Infeas. Proved	Avg Time
Plain CP-SAT	16.67% (2/12)	5/6	17.72s
+ pump hint (AddHint)	16.67% (2/12)	6/6	17.70s
linearization_level=2	8.33% (1/12)	5/6	21.22s
PORTFOLIO_SEARCH	16.67% (2/12)	5/6	19.01s
LP_SEARCH	16.67% (2/12)	5/6	18.00s

The hint doesn't find more solutions but closes an infeasibility proof, 5/6 -> 6/6. All 3 parameter tweaks were neutral or worse -- the cycle's only reproducible positive result.

Why Learning Doesn't Transfer Here

(i) Representation-level impossibility, not slow convergence — 96.8% bit-identical inputs

(ii) Exact integer feasibility is discontinuous, number-theoretic — same obstruction as parity

(iii) Market-split is built to remove smooth signal -- for LP relaxation AND gradient descent alike

Scaling depth/width/data leaves AUC at exactly 0.5 ± 0.0003 -- a hard ceiling, not a learning curve

Conclusion & Next Steps

Feasibility classification failed both our own experiments (10 results) and the literature

Learning's one positive contribution: hinting CP-SAT with symbolic byproducts, not replacing it

Classical solvers dominate through 40x100 and likely still lead at 60x150

Next: expand the hint direction (larger samples, AHL hints, partial hints); hold off on replacement